Fourier Analysis

Even & Odd Fourier Series

Even and Odd Cases of Functions

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Even and Odd Functions

How to Identify Even / Odd Functions

Even

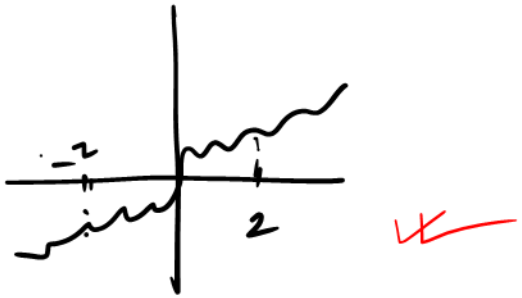
Domain $(-L, L)$

$$\boxed{f(-x) = f(x)}$$

Odd

Domain $(-L, L)$

$$\boxed{f(-x) = -f(x)}$$



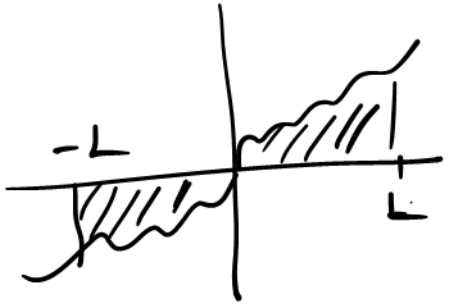
Integration of Even / Odd Functions

$f(x)$ Even

$$\int_{-L}^{L} f(x)\,dx = 2 \cdot \int_{0}^{L} f(x)\,dx$$

$f(x)$ odd

$$\int_{-L}^{L} f(x)\,dx = 0$$



Even and Odd Cases of General Fourier Series

General Fourier Series

Let $f(x)$ be defined on the interval $(-L, L)$ with period $2L$. The Fourier series expansion of $f(x)$ is defined to be

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

where the Fourier coefficients a_n and b_n are

$$\begin{cases} a_n = \dfrac{1}{L} \displaystyle\int_{-L}^{L} f(x) \cos \frac{n\pi x}{L} \, dx \,, & n = 0,1,2,3, \ldots \\ b_n = \dfrac{1}{L} \displaystyle\int_{-L}^{L} f(x) \sin \frac{n\pi x}{L} \, dx \,, & n = 1,2,3, \ldots \ldots \end{cases}$$

General Fourier Series (Even Case)

Let $f(x)$ be an **Even** function defined on the interval $(-L, L)$ with period $2L$. The Fourier series expansion of $f(x)$ is defined to be

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} \right)$$

where the Fourier coefficients a_n are

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} \, dx \,, \qquad n = 0,1,2,3, \ldots$$

General Fourier Series (Odd Case)

Let $f(x)$ be an **Odd** function defined on the interval $(-L, L)$ with period $2L$. The Fourier series expansion of $f(x)$ is defined to be

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{L} \right)$$

where the Fourier coefficients b_n are

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} \, dx \,, \qquad n = 1,2,3, \ldots$$

Problems

PROBLEM Find the Fourier series expansion of the periodic function

$$f(x) = \frac{x}{\pi}$$

over the interval $(-\pi,\ \pi)$.

$$f(-x) = \frac{-x}{\pi} = -\frac{x}{\pi} = -f(x)$$

$\Rightarrow$ odd function.

$2L = 2\pi$

$\Rightarrow L = \pi.$

$$\therefore \quad f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$= \sum_{n=1}^{\infty} b_n \cdot \sin(nx) .$$

$$\therefore \quad b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx)\, dx$$

$$\therefore\ b_n = \frac{2}{\pi}\int_0^{\pi} \frac{x}{\pi}\sin(nx)\,dx$$

$$= \frac{2}{\pi^2}\int_0^{\pi} x\cdot\sin(nx)\,dx$$

$$= \frac{2}{\pi^2}\left[x\cdot\frac{-\cos(nx)}{n} - 1\cdot\frac{-\sin(nx)}{n^2}\right]_0^{\pi}$$

$$= \frac{2}{\pi^2}\left[-x\,\frac{\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_0^{\pi}$$

x	$\sin(nx)$
1	$-\frac{\cos(nx)}{n}$
0	$-\frac{\sin(nx)}{n^2}$

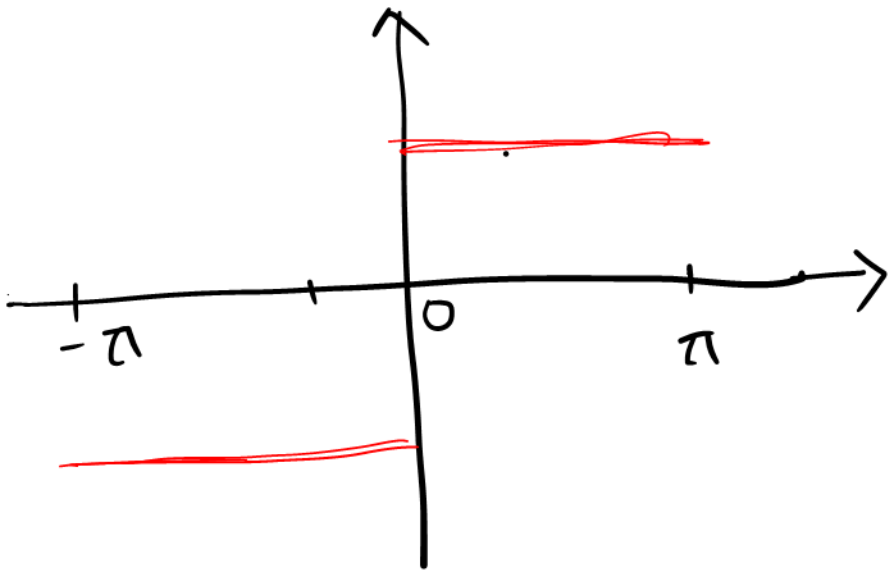
$$= \frac{2}{\pi^2}\left[\left(-\pi\,\frac{\cos(n\pi)}{n} + \cancel{\frac{\sin(n\pi)}{n^2}}\right) - \left(0\cdot\frac{1}{n} + \frac{0}{n^2}\right)\right]$$

$$= \frac{2}{\pi^2}\left[-\pi\cdot\frac{(-1)^n}{n}\right)$$

$$= \frac{-2(-1)^n}{\pi n}.$$

PROBLEM Find the Fourier series expansion of the periodic function

$$f(x) = \begin{cases} -2 & -\pi < x < 0 \\ 2 & 0 < x < \pi \end{cases}$$



$2L = 2\pi$

$\Rightarrow L = \pi.$

odd

$$\therefore \quad f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$= \sum_{n=1}^{\infty} b_n \cdot \sin(nx) .$$

$$\therefore \quad b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx)\, dx$$

$$\therefore\ b_n = \frac{2}{\pi}\int_0^{\pi} 2\sin(nx)\,dx$$

$$= \frac{4}{\pi}\int_0^{\pi} \sin(nx)\,dx$$

$$= \frac{4}{\pi}\left[\frac{-\cos(nx)}{n}\right]_0^{\pi}$$

$$= \frac{4}{\pi}\left[\frac{\cos(nx)}{n}\right]_{\pi}^{0}$$

$$= \frac{4}{\pi}\left[\frac{1}{n} - \frac{\cos(n\pi)}{n}\right]$$

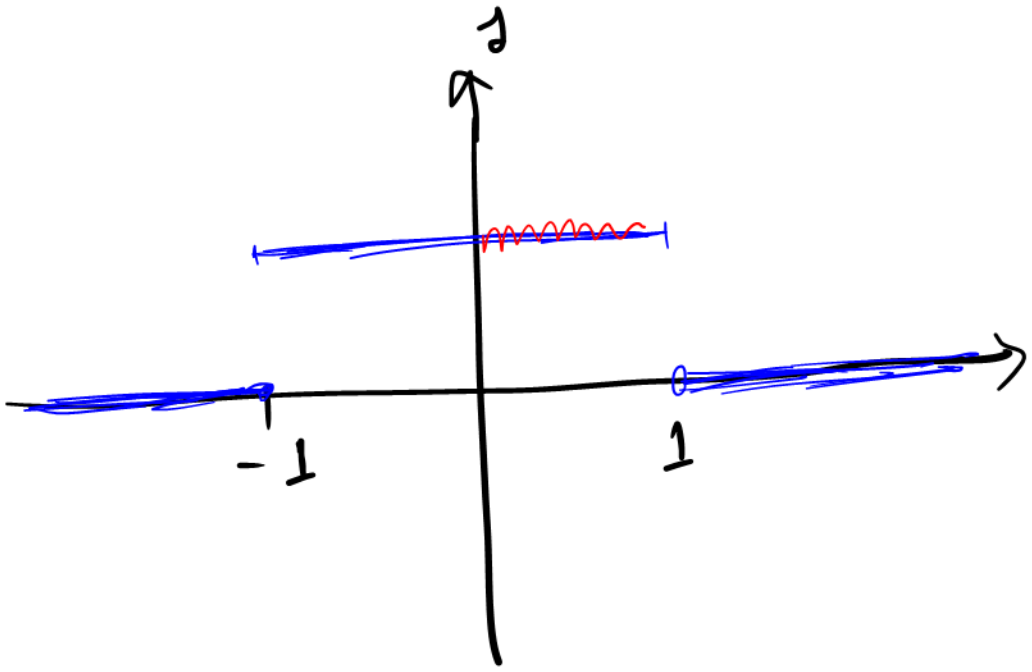
$$= \frac{4}{\pi}\,\frac{1-(-1)^n}{n}$$

$$= \frac{4\left[1-(-1)^n\right]}{n\pi}.$$

PROBLEM Find the Fourier series expansion of the periodic function $(-1, 1)$

$$f(x) = \begin{cases} 1 & |x| < 1 \\ \cancel{0} & \cancel{otherwise} \end{cases}$$

$-1 < x < 1$



Even

$2L = 2$

$\Rightarrow L = 1.$

$$\therefore \; f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

$$= \frac{a_0}{2} + \sum a_n \cos(n\pi x)$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{1}{1}\int_{-1}^{1} f(x)\cos(n\pi x)\,dx$$

(Even)

$$= \frac{2}{1}\int_{0}^{1} f(x)\cos(n\pi x)\,dx$$

$$\therefore \quad a_0 = 2 \cdot \int_0^1 f(x)\,dx$$

$$= 2 \cdot \int_0^1 1 \cdot dx$$

$$= 2 \cdot \left[x\right]_0^1$$

$$= 2.$$

$$a_n = \frac{2}{1}\int_0^1 f(x)\cos(n\pi x)\,dx$$

$$= 2\cdot\int_0^1 1\cdot\cos(n\pi x)\,dx$$

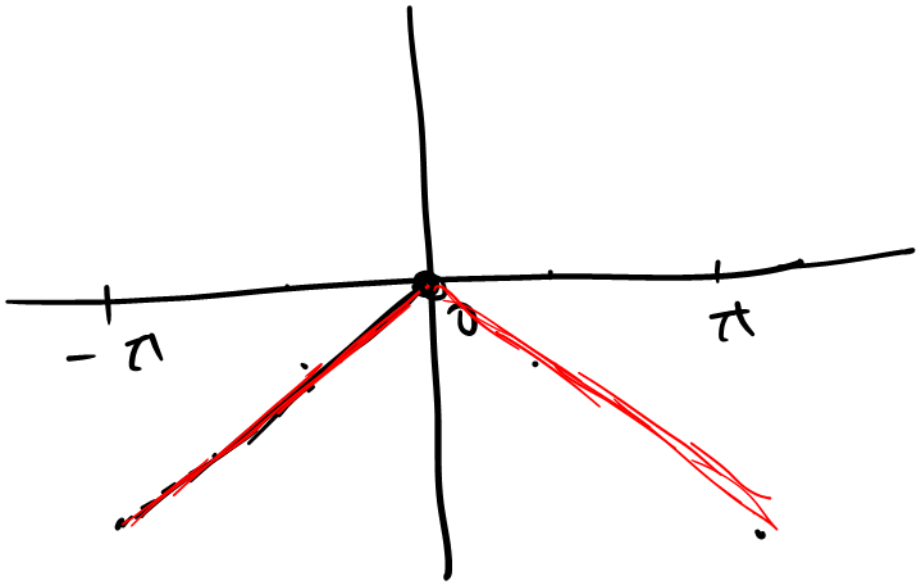
$$= 2\cdot\left[\frac{\sin(n\pi x)}{n\pi}\right]_0^1$$

$$= 2\cdot\left[\frac{\cancel{\sin}(n\pi)}{n\pi} - \frac{0}{n}\right]$$

$$= 0$$

PROBLEM Find the Fourier series expansion of the periodic function

$$f(x) = \begin{cases} x & -\pi < x < 0 \\ -x & 0 < x < \pi \end{cases}$$



Even

$2L = 2\pi$

$\Rightarrow L = \pi$

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx)\, dx$$

$$\therefore \quad a_0 = \frac{2}{\pi} \int_0^{\pi} f(x)\, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (-x)\, dx$$

$$= \frac{-2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi}$$

$$= \frac{-2}{\pi} \cdot \frac{\pi^2}{2} = -\pi.$$

$$\therefore \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx)\, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (-x) \cos(nx)\, dx$$

$$= \frac{2}{\pi} \left[(-x) \cdot \frac{\sin(nx)}{n} - (-1) \frac{-\cos(nx)}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-x \frac{\sin(nx)}{n} - \frac{\cos(nx)}{n^2} \right]_0^{\pi}$$

$-x$	$\cos(nx)$
-1	$\frac{\sin(nx)}{n}$
0	$-\frac{\cos(nx)}{n^2}$

$$= \frac{2}{\pi}\left[x\,\frac{\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]_{\pi}^{0}$$

$$= \frac{2}{\pi}\left[\left(0 + \frac{1}{n^2}\right) - \left(\pi \cdot \cancel{\frac{\sin(n\pi)}{n}} + \frac{\cos(n\pi)}{n^2}\right)\right]$$

$$= \frac{2}{\pi}\left[\frac{1}{n^2} - \frac{(-1)^n}{n^2}\right]$$

$$= \frac{2\cdot\left(1-(-1)^n\right)}{\pi n^2}.$$

PROBLEM Find the Fourier series expansion of the periodic function

$$f(x) = |x| \quad -\pi < x < \pi$$

$$f(-x) = |-x|$$

$$= |x|$$

$$= f(x)$$

Even.

$$2L = 2\pi$$

$$\Rightarrow L = \pi.$$

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx)\, dx$$

$$\therefore \quad a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} |x|\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x\,dx$$

$$= \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \frac{2}{\pi}\cdot\left(\frac{\pi^2}{2} - 0\right) = \pi.$$

$* \; |x| = x, \quad x \geq 0$

$* \; |x| = -x, \quad x < 0$

$$\therefore a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} |x|\cdot\cos(nx)\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x\cdot\cos(nx)\,dx$$

$$= \frac{2}{\pi}\left[x\cdot\frac{\sin(nx)}{n} - 1\cdot\frac{-\cos(nx)}{n^2}\right]_0^{\pi}$$

x	$\cos(nx)$
1	$\frac{\sin(nx)}{n}$
0	$\frac{-\cos(nx)}{n^2}$

$$= \frac{2}{\pi}\left[x \cdot \frac{\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\left(\pi \cdot \cancel{\frac{\sin n\pi}{n}} + \frac{\cos(n\pi)}{n^2}\right) - \left(0 + \frac{1}{n^2}\right)\right]$$

$$= \frac{2}{\pi}\left[\frac{(-1)^n}{n^2} - \frac{1}{n^2}\right).$$

PROBLEM Find the Fourier series expansion of the periodic function

$$f(x) = x^2$$

over the interval $(-3,\ 3)$.

$f(-x)$

$= (-x)^2$

$= x^2 = f(x)$

Even

$2L = 6$

$\Rightarrow L = 3.$

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{3}\right).$$

$$a_n = \frac{2}{3} \int_0^3 f(x) \cos\left(\frac{n\pi x}{3}\right) dx$$

$$\therefore \quad a_0 = \frac{2}{3} \int_0^3 f(x)\, dx$$

$$= \frac{2}{3} \int_0^3 x^2\, dx$$

$$= \frac{2}{3} \cdot \left[\frac{x^3}{3} \right]_0^3$$

$$= \frac{2}{3} \cdot \left[\frac{27}{3} \right] = 6.$$

$$\therefore \quad a_n = \frac{2}{3}\int_0^3 f(x)\cos\left(\frac{n\pi x}{3}\right)dx$$

$$= \frac{2}{3}\int_0^3 x^2 \cos\left(\frac{n\pi x}{3}\right)dx$$

$$= \frac{2}{3}\left[x^2 \cdot \frac{\sin\left(\frac{n\pi x}{3}\right)}{\frac{n\pi}{3}} - 2x \cdot \frac{-\cos\left(\frac{n\pi x}{3}\right)}{\frac{n^2\pi^2}{9}} + 2 \cdot \frac{-\sin\left(\frac{n\pi x}{3}\right)}{\frac{n^3\pi^3}{27}}\right]_0^3$$

x^2	$\cos\left(\frac{n\pi x}{3}\right)$
$2x$	$\frac{\sin\left(\frac{n\pi x}{3}\right)}{\frac{n\pi}{3}}$
2	$\frac{-\cos\left(\frac{n\pi x}{3}\right)}{\frac{n^2\pi^2}{9}}$
0	$\frac{-\sin\left(\frac{n\pi x}{3}\right)}{\frac{n^3\pi^3}{27}}$

$$= \frac{2}{3}\left[\frac{3x^2}{n\pi}\sin\left(\frac{n\pi x}{3}\right) + \frac{18x}{n^2\pi^2}\cos\left(\frac{n\pi x}{3}\right) - \frac{54}{n^3\pi^3}\sin\left(\frac{n\pi x}{3}\right)\right]_0^3$$

$$= \left[\frac{2x^2}{n\pi}\sin\left(\frac{n\pi x}{3}\right) + \frac{12x}{n^2\pi^2}\cos\left(\frac{n\pi x}{3}\right) - \frac{36}{n^3\pi^3}\sin\left(\frac{n\pi x}{3}\right)\right]_0^3$$

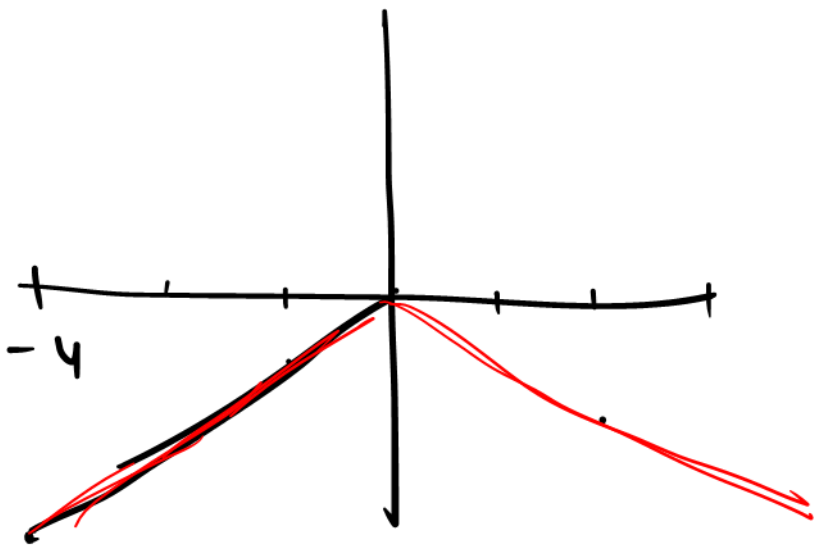
$$= \left(\frac{2\cdot 3^2}{n\pi}\cancel{\sin(n\pi)} + \frac{12\cdot 3}{n^2\pi^2}\cdot\cos(n\pi) - \frac{36}{n^3\pi^3}\cdot\cancel{\sin(n\pi)}\right)$$

$$- \left(0 + 0 - 0\right)$$

$$= \frac{36}{n^2\pi^2} \cdot (-1)^n .$$

PROBLEM Find the Fourier series expansion of the periodic function

$$f(x) = \begin{cases} x & -4 < x < 0 \\ -x & 0 < x < 4 \end{cases}$$



Even.

$2L = 8$

$\Rightarrow L = 4$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{4}\right)$$

$$a_n = \frac{2}{4} \int_0^4 f(x) \cos\left(\frac{n\pi x}{4}\right) dx$$

$$a_0 = \frac{2}{4}\int_0^4 f(x)\,dx$$

$$= \frac{1}{2}\int_0^4 (-x)\,dx$$

$$= \frac{1}{2}\left[-\frac{x^2}{2}\right]_0^4$$

$$= \frac{1}{4}\left[x^2\right]_4^0 = \frac{1}{4}(0-16) = -4$$

$$\therefore \quad a_n = \frac{2}{4}\int_0^4 f(x)\cos\left(\frac{n\pi x}{4}\right)dx$$

$$= \frac{1}{2}\int_0^4 (-x)\cos\left(\frac{n\pi x}{4}\right)dx$$

$$= \frac{1}{2}\left[(-x)\frac{\sin\left(\frac{n\pi x}{4}\right)}{\frac{n\pi}{4}} - (-1)\frac{-\cos\left(\frac{n\pi x}{4}\right)}{\frac{n^2\pi^2}{16}}\right]_0^4$$

$-x$	$\cos\left(\frac{n\pi x}{4}\right)$
-1	$\frac{\sin\left(\frac{n\pi x}{4}\right)}{\frac{n\pi}{4}}$
0	$\frac{-\cos\left(\frac{n\pi x}{4}\right)}{\frac{n^2\pi^2}{4^2}}$

$$= \left[\frac{-2x}{n\pi} \sin\left(\frac{n\pi x}{4}\right) - \frac{8}{n^2\pi^2} \cos\left(\frac{n\pi x}{4}\right) \right]_0^4$$

$$= \left[\frac{2x}{n\pi} \sin\left(\frac{n\pi x}{4}\right) + \frac{8}{n^2\pi^2} \cos\left(\frac{n\pi x}{4}\right) \right]_4^0$$

$$= \left(0 + \frac{8}{n^2\pi^2} \cdot 1 \right) - \left(\frac{8}{n\pi} \sin(n\pi) + \frac{8}{n^2\pi^2} \cos(n\pi) \right)$$

$$= \frac{8}{n^2\pi^2} - \frac{8(-1)^n}{n^2\pi^2}$$

$$= \frac{8 \cdot \left(1 - (-1)^n\right)}{n^2 \pi^2}$$

Thank you!